

12.116

AI25BTECH11012 - GARIGE UNNATHI

Question:

Let $\mathbf{P}$ and $\mathbf{Q}$ be two square matrices of same size. Consider the following statements

- 1) $\mathbf{PQ} = 0$ then $\mathbf{P}=0$ or $\mathbf{Q} =0$ or both
- 2) $\mathbf{PQ}=\mathbf{I}$ then $\mathbf{QP} = \mathbf{I}$
- 3) $(\mathbf{P} + \mathbf{Q})^2 = \mathbf{P}^2 + 2\mathbf{PQ} + \mathbf{Q}^2$
- 4) $(\mathbf{P} - \mathbf{Q})^2 = \mathbf{P}^2 - 2\mathbf{PQ} + \mathbf{Q}^2$

Determine which statements are True and which are False

Solution:

- 1) This option is False

This can be proven by a simple counter example :

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \mathbf{Q} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad (1.1)$$

$$\mathbf{PQ} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad (1.2)$$

- 2) Given $\mathbf{PQ} = \mathbf{I}$

From this we can say that $|\mathbf{P}|$ and $|\mathbf{Q}|$ is not zero

That is inverse exists

$$\mathbf{Q} = \mathbf{P}^{-1} \quad (2.1)$$

$$\mathbf{QP} = \mathbf{P}^{-1}\mathbf{P} = \mathbf{I} \quad (2.2)$$

Hence this option is true

- 3)

$$(\mathbf{P} + \mathbf{Q})^2 = (\mathbf{P} + \mathbf{Q})(\mathbf{P} + \mathbf{Q}) \quad (3.1)$$

$$= \mathbf{P}^2 + \mathbf{PQ} + \mathbf{QP} + \mathbf{Q}^2 \quad (3.2)$$

This equation will not be equal to $\mathbf{P}^2 + 2\mathbf{PQ} + \mathbf{Q}^2$

this is because Matrix multiplication is not commutative

$$\mathbf{PQ} \neq \mathbf{QP} \quad (3.3)$$

this option is false

- 4) similarly option 4 is also False

Hence only option 2 is true